
Simulation of Flow and Transport in the Partially Molten Mantle of the Earth

Chenyu Tian

Oden Institute, University of Texas at Austin

Todd Arbogast, Oden Institute & Mathematics, University of Texas at Austin
Marc A. Hesse, Oden Institute & Geosciences, University of Texas at Austin



Motivation

Model relative motion between solid and liquid.

Deal with zero porosity (no melting) region mathematically sound.

Transport capturing discontinuity.



McKenzie Equations

conservation laws and compaction equation



Alternative Formulation

Spiegelman.

Aspect

Rhoberg

CitcomS



Two options

melting function

Phase calculation





Numerical framework





$H(\text{div})$ - and H^1 - conforming



